SHEHBAZ PATHAN

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SUMMARY

AI Engineer shipping production LLM agents at Opentrons: a simulator-validated assistant that turns scientists' plain-English requests into robot protocols. Builds agentic systems, RAG, and LLM evals; M.S. Data Analytics (4.0 GPA), research on agent evaluation.

SKILLS

LLMs / Agents: Agentic AI, LangGraph, LangChain, RAG, Vector Search, Reranking, LLM Evals, Speculative Decoding, Fine-Tuning
ML / DL: PyTorch, Hugging Face, FSDP/DDP, PINNs, Neural Operators (FNO), TensorFlow, Scikit-learn, XGBoost, LightGBM
Infra / MLOps: Python, SQL, TypeScript, FastAPI, Docker, Kubernetes, AWS (S3, SageMaker), MLflow, CI/CD (GitHub Actions)
Data / BI: PostgreSQL, ChromaDB, MongoDB, Tableau, Power BI, DAX, Statistical Modeling, Time-Series Forecasting, A/B Testing

EXPERIENCE

AI Engineer, Opentrons Labworks Inc. | New York, NY — Aug 2026 – Present

- Shipped a production LLM assistant (OpenRouter, FastAPI, Docker) that turns scientists' plain-English lab methods into runnable Opentrons Protocol API code for OT-2 and Flex robots, asking clarifying questions instead of guessing missing details.
- Built guardrails against hallucinated hardware: grounded the LLM in Opentrons' official labware and pipette catalog, constrained output to a structured JSON plan schema covering 13 liquid-handling commands, and released only simulator-verified code.
- Made generation self-correcting and resilient with an agentic generate-simulate-repair loop (up to 3 attempts) and a deterministic fallback when the LLM is down; regression-gated by 82 tests, including 5 golden natural-language evals run in the simulator.

AI Automation Engineer Intern, Rocket Autoparts | Albuquerque, NM (Remote) — Feb 2026 – May 2026

- Replaced manual handoffs in 3 core workflows (supply chain, inventory, logistics) with LLM-integrated automation.
- Cut order-error rates 15% across tracked SKUs by deploying ML forecasting and anomaly detection on fulfillment bottlenecks.
- Reduced cross-team reporting delays by parsing unstructured emails and order forms into structured records for decision engines.

Data Analyst, Automation Intern, BTS | New York, NY — Jun 2025 – Aug 2025

- Cut manual processing time ~40% and report turnaround from 2 days to under 4 hours by automating Salesforce reporting in Python; built CRM ingestion, deduplication, and enrichment pipelines for 5+ client accounts.
- Delivered Power BI dashboards on 15+ KPIs, adopted as the permanent leadership reporting layer for 3 consulting engagements.

Data Analyst, Dreamcare Developers Pvt. Ltd. | Pune, India — Jun 2022 – Jan 2024

- Guided roadmap priorities over 3 product cycles with regression and significance tests on 500K+ row datasets for 10+ clients.
- Saved ~8 analyst-hours/week by automating reporting; built Tableau/Power BI dashboards adopted as the single source of truth.

PROJECTS

spectrace | *Python, CLI, pytest, GitHub Actions, MIT* — github.com/shehbaz0101/spectrace

- Showed speculative-decoding metrics miss bad agent steps: on 15 agent traces (49 steps), a step-level verifier (TypeSafe Jev) rejected all 11 harmful steps (e.g., DROP TABLE, nonexistent tools) that token acceptance would have kept.
- Released an open-source, $0-by-default harness that replays multi-step tool trajectories and reports latency, draft accept rate, prefill token growth, and cost per successful task; 77 keyless pytest tests and an offline demo run in CI.

PINNForge | *PyTorch, FastAPI, Pydantic* — github.com/shehbaz0101/pinnforge

- Recovered Burgers viscosity from 32 sparse sensors at 3.4% mean error on hard OOD cases vs. 35% for a closed-form estimator (which failed 9 of 30) via preregistered, five-seed Fourier neural operator studies.
- Built a PINN sandbox (harmonic, Burgers, Poisson) with a config-driven CLI and rate-limited FastAPI service; released v0.2.2.

Coticula | *PyTorch, NumPy, SciPy* — github.com/shehbaz0101/coticula

- Built a four-exam benchmark (Predict, Conserve, Counterfactual, Explain) for PDE models (FNO/PINO) on 1D Burgers and 2D heat; released v0.3.2 with a 48-item rule-graded exam, a grade-your-own-model CLI, and a citation file.
- Showed FNO beats PINO on L2 error but trails it on PDE residual, and that FNO barely reacts to a 1.5x diffusivity change; added fail-closed OOD and untrusted flags.

SmartBite | *FastAPI, XGBoost, LightGBM*

- Achieved 87% precision across 5 violation categories at sub-200ms latency in a NYC health-inspection risk platform.

EDUCATION

M.S. in Data Analytics and Visualization, Yeshiva University | New York, NY — May 2026
GPA 4.0 / 4.0 • Coursework: Machine Learning, Deep Learning, NLP, Time-Series, Data Engineering

B.S. in Computer Applications, BAMU University | Pune, India — May 2022
GPA 3.8 / 4.0

PUBLICATIONS

The Illusion of Success: Trajectory-Level Evaluation for Enterprise Workflow Agents. Under Review, 2025.